



Francesco Di Santo

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Gender: Male **Date of birth**: 19/04/2001 **Place of birth**: Atesa (CH), Italy **Nationality**: Italian

ABOUT ME

Computer Engineering graduate specializing in Computer Graphics and Multimedia. Solid experience in Extended Reality (XR), 3D Modeling and Animation, and Human-Computer Interaction (HCI). Expertise in developing immersive applications and intelligent systems using Computer Vision and Machine Learning techniques. Experience in Unity, Blender, C#, and Python, with a proactive and solution-oriented mindset.

EDUCATION AND TRAINING

[10/2023 – Current]

Master's Degree in Computer Engineering

Politecnico di Torino <https://www.polito.it/>

City: Torino | **Country**: Italy | **Field(s) of study**: Graphics and Multimedia | **Level in EQF**: EQF level 7 | **Type of credits**: CFU | **Number of credits**: 120 | **Thesis**: Cognitive state assessment in extended reality environments using active and passive brain-computer interfaces

- **Virtual and Augmented Reality**: Design of immersive environments (XR/VR/AR) and development of interactive 3D applications using Unity.
- **Computer Graphics and Computer Vision**: Advanced study of rendering algorithms, geometric modeling, and computer vision techniques for pattern recognition. **Multimedia Systems**: Analysis, processing, and compression of audio/video signals; study of multimedia architectures and data transmission.
- **Human-Computer Interaction (HCI)**: User-centered interface design, usability testing, and interaction methodologies.
- **Game and Application Development**: Application development techniques and game design principles.

[09/2020 – 09/2023]

Bachelor's Degree in Computer Engineering

Politecnico di Torino <https://www.polito.it/>

City: Torino | **Country**: Italy | **Field(s) of study**: Information and Communication Technologies | **Final grade**: 102/110 | **Level in EQF**: EQF level 6 | **Type of credits**: CFU | **Number of credits**: 182

- **ICT Engineering and Problem Solving**: Innovation and application of ICT tools through an engineering approach to address challenges across a broad spectrum of domain applications.
- **Computer Architecture**: Understanding of fundamental principles of computer architecture and the hardware-software integration in modern computing systems.
- **Software Proficiency**: Knowledge of operating systems, programming languages (including object-oriented programming), and software engineering methodologies.
- **Data Management**: Principles and technologies for database modeling, design, and management.
- **Fundamental Sciences**: Core knowledge in mathematics, physics, and electronics (applied electronics, circuits, and systems).

[09/2015 – 06/2020]

Scientific High School Diploma

WORK EXPERIENCE

EvoHunt S.R.L. <https://evohunt.com/>

City: Torino | **Country:** Italy | **Business or sector:** Information and communication

[04/2023 – 08/2023] **Curricular internship**

This project focused on the front-end development of a new version of an HR Tech web application, which automates the candidate recruitment process by integrating profiling tools and machine learning models. The work centered on designing, managing, and optimizing the user interface (User Experience) using modern technologies such as React, TypeScript, JavaScript, Formik, and Bootstrap. Key responsibilities included implementing dynamic forms and interactive components with state management and modals, simulating API integration with the back-end using dummy data, and adopting a collaborative workflow based on Git and Node.js.

Politecnico di Torino

City: Torino | **Country:** Italy

[03/2025 – 09/2025] **Student collaboration**

Support activities for the organization of graduation ceremonies: reception/information desk/material preparation at the Engineering Careers Office.

Fondo Ambiente Italiano

City: Atessa (CH) | **Country:** Italy

[03/2018] **Work-study program (PCTO)**

Cultural guide and management of tourist flows for the enhancement of local historical heritage, with detailed illustrations of exclusive sites.

PROJECTS

[03/2025 – 09/2025] **Song noise reduction without clean training data**

In this project, a deep learning-based audio denoising pipeline was developed using a 20-layer Deep Complex U-Net (DCUNet-20) architecture to perform music denoising without requiring clean training data. Adopting the Noise2Noise paradigm, the network was trained exclusively on pairs of noisy audio signals generated by combining tracks from the MUSDB18 dataset with environmental and gaussian sound events from UrbanSound8K. The implementation operates directly on complex-valued Short-Time Fourier Transform (STFT) spectrograms to preserve both magnitude and phase information. Key technical highlights include complex convolutions, complex batch normalization, complex weight initialization based on Rayleigh distribution, and gradient optimization using a weighted Signal-to-Distortion Ratio (wSDR) loss. The final optimized system achieved an average Signal-to-Noise Ratio (SNR) improvement of +7.54 dB, outperforming baseline speech-oriented models on musical signals.

Link: <https://github.com/Frads01/mlvm-noise-remover>

[06/2025 – 07/2025] **TaskLass: Full-Stack Web Platform for Managing and Grading School Assignments**

TaskLass is a full-stack web app developed for the Web Applications 1 exam, designed to digitize the assignment, submission, and grading of coursework. The application supports two distinct user roles: teachers, who can create assignments for specific student groups

and grade submitted work, and students, who can view assignments, submit responses, and track their grade point average calculated server-side. The architecture features a reactive frontend built with React and React Router, utilizing modular components for separate dashboards, paired with a Node.js/Express backend that exposes REST APIs with session-based authentication and structured HTTP error handling. Data is stored in a relational database configured with many-to-many relationships. Key skills gained during development include relational schema modeling, designing authenticated REST APIs, managing client concurrency, and structuring a component-based user interface.

Link: <https://github.com/Frads01/progetto-webapp1>

[05/2025 – 07/2025] **Face Verification: geometric landmarks and neural embeddings comparison**

This project evaluates two facial verification methods built into an iOS application using Unity, C#, and AR Foundation: a 3D Geometric Approach and a 2D Neural Approach. The 3D approach captures TrueDepth point clouds via ARKit to measure normalized Euclidean distances across 17 anatomical landmarks (from 1,220 tracked points). The 2D approach uses ArcFace (ResNet-100 backbone) on Unity's Barracuda engine to compare 512-dimensional embeddings from RGB images using cosine similarity. The application features real-time face detection, a 3D facial viewer, live dual-pipeline testing, biometric measurement logging, and CSV export. Experimental results across 100+ tests showed the 3D method achieved 97.12% overall accuracy and 100% recall, whereas the ArcFace model achieved 100% precision.

Link: <https://github.com/SalvatoreGiugliano98/FaceVerification>

[11/2024 – 02/2025] **DriveAcademyVR**

Drive Academy VR is a virtual reality driving school simulator developed for the Meta Quest 3 headset. It provides an immersive, risk-free environment for users to practice driving, learn traffic rules, and receive real-time vocal guidance from a virtual instructor. Built with realistic vehicle physics, the system features a fully interactive cockpit where users manually operate controls such as the steering wheel, gear shift, turn signals, handbrake, and seatbelt, backed by a diegetic UI integrated into the dashboard. Operating within a grid-based city filled with traffic and pedestrians, the application tracks performance, provides immediate visual, auditory, and haptic feedback on infractions, and includes an introductory tutorial module.

Links: <https://frads01.itch.io/drive-academy> | <https://github.com/SalvatoreGiugliano98/DriveAcademyVR>

[10/2024 – 02/2025] **My Museum Adventure**

Developed for the Human-Computer Interaction course at Politecnico di Torino, *My Museum Adventure* is an Android application designed to gamify museum visits through theme-based interactive quests. Users select narrative paths (e.g., Adventure, Fantasy, Sci-fi) and solve riddles to discover artworks, earning medals and achievements along the way. Key features include camera-based artwork scanning via external image recognition APIs to verify answers, crowdsourced interactive maps for navigation, and detailed descriptions with multilingual audio guides. Technically, the app is built with Kotlin and Jetpack Compose, integrating Room for local offline storage, CameraX, OkHttp for API calls, and Jetpack Navigation.

Link: <https://github.com/SalvatoreGiugliano98/my-museum-adventure-main>

[10/2024 – 01/2025] **Multi-platform Screen-casting**

Developed for the System Programming course at Politecnico di Torino, this cross-platform PC application written in Rust enables real-time screen sharing on Windows, macOS, and Linux using IPv4 multicast. Built with an intuitive egui graphical interface, it supports two main roles:

- **Sender:** Configures IP/port and monitor selection, with real-time controls to pause sharing, send a blank white screen, draw annotations (rectangles, arrows), share a specific screen region, or stop the stream (supported by hotkeys like Ctrl+P, Ctrl+B, Ctrl+D, Ctrl+S, Ctrl+Q). Includes fallback handling if a monitor disconnects.
- **Receiver:** Connects via IPv4 address and port, featuring session exit options and built-in screen recording powered by ffmpeg (with automatic download handling if not pre-installed).

Link: <https://github.com/Frads01/pds-screencast>

[05/2024 – 10/2024] **Canoa Slalom**

This project was developed for the Image Processing and Computer Vision course at Politecnico di Torino to analyze Canoe Slalom performances. Built using Python and OpenCV, the system applies computer vision techniques to track the canoe and detect slalom gates in dynamic video streams, addressing challenges like water reflections and camera movement. Its key features include gate detection and segmentation, athlete and boat tracking, automated evaluation of gate touches or missed gates, and optimized frame processing. By leveraging techniques like morphological operations, contour detection, and optical flow, the project applies image processing theory to real-world sports analytics.

Link: <https://github.com/Frads01/ipcv-canoa>

[10/2023 – 06/2024] **Computer Graphics: 3D modeling and interactive graphics application**

This project was developed for a Computer Graphics course to implement the core principles of 3D rendering, modeling, and interactive visual computing. It consists of a real-time, interactive 3D application built with low-level graphics libraries like OpenGL, along with a 3D scene modeled in Blender using physics techniques. The application features a navigable 3D environment with camera controls, geometric transformations, scene graph management, dynamic lighting (Phong/Blinn-Phong), texture mapping, and custom shaders, bridging mathematical concepts with practical graphics programming.

Link: <https://github.com/SalvatoreGiugliano98/Computer-Graphics>

[01/2024 – 02/2024] **Quoridor**

Quoridor is a from-scratch embedded build of the classic board game, developed for the Computer Systems Architecture course at Politecnico di Torino on an ARM Cortex-M3 platform (Keil µVision), written in C with critical routines in Assembly. The 13×13 board is modeled as a matrix alternating player and wall cells, with a recursive DFS algorithm ensuring no wall can ever fully trap an opponent. Interaction runs through a GLCD display and touch panel for rendering, a joystick and buttons (RIT-based polling and debouncing) for input, and several hardware timers managing the 20-second move clock and on-screen messages. A single-board mode features an NPC that pathfinds via a Dijkstra-based algorithm, while a two-board mode enables CAN bus multiplayer, synchronized through a handshake protocol.

Link: <https://github.com/Frads01/ase-quoridor>

SKILLS

computer technology | human-computer interaction | algorithms | multimedia systems | mathematics | computer graphics | virtual reality | augmented reality | computer vision

Soft skills

attend to detail | think creatively | logic | troubleshoot | adapt to change | Operational autonomy and ability to work in a team | manage personal progression

Software e linguaggi di programmazione

Python (computer programming) | GIMP (graphics editor software) | Unity (digital game creation systems) | graphics editor software | C#

CERTIFICATIONS

[Cambridge Assessment English, 11/07/2018]

Cambridge English Level 1 Certificate in ESOL International (First) - Score 166

Mode of learning: Presential

[IDCERT, 05/04/2025]

DigComp 2.2 Highly Specialized 8

Mode of learning: Online

Link: <https://it.idcert.io/idfy/LUZQXC1NNE>

LANGUAGE SKILLS

Mother tongue(s): Italiano

Other language(s):

English

LISTENING B2 READING B2 WRITING B2

SPOKEN PRODUCTION B2 SPOKEN INTERACTION B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

VOLUNTEERING

[08/2018]

Guide for FAI "Arte e Cultura" Days Atessa (CH), Italy

Served as a cultural guide during the "Arte e Cultura" days organized by Fondo Ambiente Italiano (FAI), contributing to the enhancement and promotion of local artistic heritage through guided tours and public engagement.

[2010 – 2020]

AGESCI (Italian Catholic Guides and Scouts Association) Atessa (CH), Italy

Active participation in community service activities, teamwork, and outdoor experiences. Development of a strong ability to adapt, management of volunteer projects, and the ability to collaborate in dynamic environments.